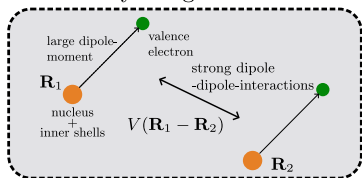


Rydberg atoms



$$H_{\text{eff}} = \sum_{\substack{i,j=1 \\ c,d \in \{a,b\}}}^{N/2} U_{i,j}^{c,d} c_i^\dagger d_j \quad \text{for } i \neq j: U_{ij}^{a,b} = \frac{1}{m\omega} \frac{\partial^2 V(\mathbf{R}_i^A - \mathbf{R}_j^B)}{\partial \delta \mathbf{R}_i^A \partial \delta \mathbf{R}_j^B}$$

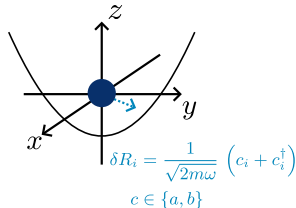
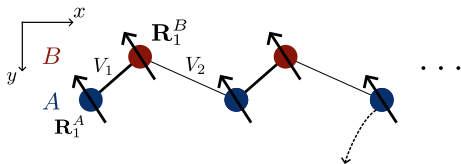
imposing **chiral symmetry**:
sublattices (●,●) can not
interact with each other

$$H_{\text{eff}} = \sum_{i,j} U_{i,j}^{a,b} a_i^\dagger b_j + \text{h.c.}$$

1D phonons:

can be mapped to the
extended ssh model

preparing the atoms in the Rydberg states:



2D phonons:

two independent nearest neighbor
ssh models